

MINI-PROJECT REPORT

Smart Street Light System

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ABSTRACT

The Smart Street Light System with Vehicle Detection and LCD Monitoring is an Arduino UNO-based automation project designed to improve energy efficiency and enhance road safety. The system automatically controls street lights based on ambient light conditions and vehicle presence.

An LDR (Light Dependent Resistor) detects day and night conditions, while an HC-SR04 ultrasonic sensor detects vehicles or pedestrians approaching the illuminated area. During daytime, all street lights remain switched OFF to conserve energy. During nighttime, two base street lights remain ON to provide minimum illumination. When a vehicle is detected within a predefined distance, additional street lights are activated instantly.

A 16×2 LCD display continuously monitors and displays the current operating mode and vehicle detection status. The system serves as a low-cost prototype of modern intelligent street lighting solutions used in smart cities.

1. Introduction

Street lighting is an essential part of modern transportation infrastructure. Conventional street lighting systems operate continuously throughout the night regardless of traffic conditions, resulting in significant energy wastage and increased operational costs. With the advancement of embedded systems and automation technologies, intelligent street lighting systems have become an effective solution for reducing power consumption while maintaining road safety.

The Smart Street Light System is designed to automate street light operation based on environmental conditions and vehicle movement. The Arduino UNO acts as the central controller that processes sensor inputs and controls the street lights accordingly.

The system follows these operational strategies:

- During daytime, all street lights remain OFF
- During nighttime, two base street lights remain ON
- Additional street lights are activated when a vehicle is detected
- The additional lights remain ON for a short duration after the vehicle passes
- LCD monitoring provides real-time system status

2. Objectives

The main objectives of this project are:

1. Design and implement a smart street lighting system using Arduino UNO.
2. Automatically detect day and night conditions using an LDR sensor.
3. Detect vehicles using an HC-SR04 ultrasonic sensor.
4. Reduce energy consumption by activating lights only when required.
5. Implement intelligent street light control logic.
6. Provide real-time system monitoring using a 16×2 LCD display.
7. Simulate multiple street lights using LEDs.
8. Gain practical knowledge of embedded systems and sensor integration.
9. Develop a low-cost prototype suitable for smart city applications.

3. Hardware Components

3.1 Arduino UNO

The Arduino UNO serves as the main controller of the system. It reads sensor inputs, processes the control logic, updates the LCD display, and controls the LEDs representing street lights.

- ATmega328P microcontroller
- 14 Digital I/O Pins and 6 Analog Inputs
- Easy programming using Arduino IDE
- Low cost and wide availability

3.2 LDR Sensor Module

The LDR sensor detects ambient light intensity and determines day/night conditions, providing input to the Arduino for automatic operation and enabling energy-efficient street light control.

3.3 HC-SR04 Ultrasonic Sensor

The HC-SR04 sensor measures distance using ultrasonic waves to detect approaching vehicles and triggers additional street lights when a vehicle is present within the predefined range.

3.4 16×2 LCD Display

The LCD provides real-time information about system operation — displaying current mode (Day/Night), vehicle detection status, and system monitoring information.

3.5 LEDs and Other Components

Five LEDs simulate street lights on the road. 220Ω resistors are used for current limiting of LEDs. A 10kΩ resistor is used in the LDR voltage divider circuit. A breadboard and jumper wires are used for circuit assembly.

4. Working Principle

4.1 Day/Night Detection

The LDR continuously measures ambient light intensity. During daytime, the LDR detects sufficient light and the Arduino switches OFF all street lights to minimize energy consumption. When darkness is detected, the system enters Night Mode.

4.2 Night Mode Operation

When darkness is detected, two base street lights remain ON continuously to provide minimum road illumination. The ultrasonic sensor becomes active during night mode to monitor approaching vehicles.

4.3 Vehicle Detection

The HC-SR04 continuously monitors the road. When a vehicle enters the predefined detection range, additional street lights are activated immediately, increasing road illumination for improved safety.

4.4 Delay Function

After the vehicle leaves the detection range, additional lights remain ON for 3 seconds before turning off. This prevents sudden switching and achieves realistic street lighting behavior similar to real-world systems.

4.5 LCD Monitoring

The LCD continuously displays system status such as Mode: DAY / Lights: OFF or Mode: NIGHT / Vehicle: YES. This provides real-time monitoring without requiring a computer connection.

5.Circuit Description

5.1 LDR Sensor Connections

LDR Module Pin	Arduino UNO Pin
VCC	5V
GND	GND
D0	D2

5.2 HC-SR04 Ultrasonic Sensor Connections

HC-SR04 Pin	Arduino UNO Pin
VCC	5V
GND	GND
TRIG	D7
ECHO	D6

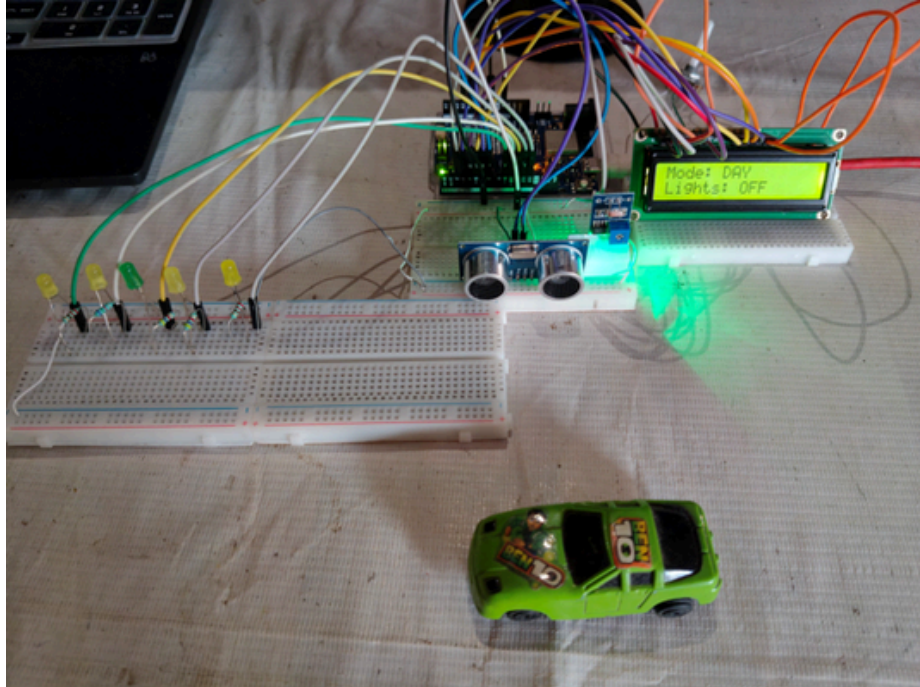
5.3 LED Connections

LED	Arduino Pin	Resistor
LED1	D9	220 Ohm
LED2	D10	220 Ohm
LED3	D11	220 Ohm
LED4	D12	220 Ohm
LED5	D13	220 Ohm

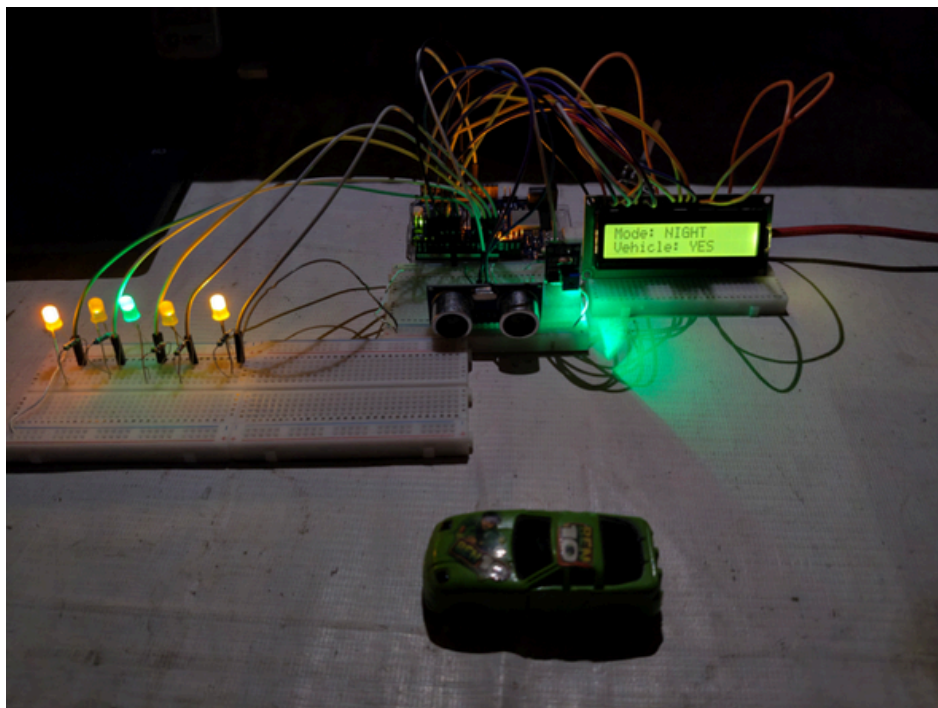
5.4 LCD Display Connections

LCD Pin	Arduino UNO Pin
RS	D3
E	D4
D4	D5
D5	A0
D6	A1

6. Setup



Day Mode (All Lights OFF)



Night Mode (All Lights ON)

7. Test Results

7.1 Functional Test Summary

The system was tested across all three time modes with various increment settings. Results confirmed accurate timing and reliable switching in all cases.

Test Conditions	Expected Result	Actual Result
Daytime (bright light)	All lights off	Pass
Nighttime (dark)	base lights ON (2 LEDs)	Pass
Vehicle detected (night)	All lights ON (5 LEDs)	Pass
Vehicle Leaves	Lights ON for 3 sec	Pass
LCD Monitoring	Status Updated	Pass
Day to Night Transition	Smooth mode switching	Pass

7.2 Observations

- Day/Night detection was accurate and reliable.
- Vehicle detection was responsive with no significant delay.
- LCD updated correctly with each state change.
- No false triggering was observed during testing.
- Delay functionality worked reliably — lights stayed ON for 3 seconds post-detection.
- Overall system performance was stable under all test conditions.

8.Applications

8.1 Smart Cities

The system can be deployed for intelligent street lighting in urban smart city infrastructure, enabling centralized monitoring and control of road lighting.

8.2 Highway Lighting

Reduces power consumption on roads with low traffic by activating full lighting only when vehicles are present.

8.3 Parking Areas

Provides automatic lighting when vehicles enter parking lots or driveways, improving safety and saving energy.

8.4 Residential Streets

Improves safety while minimizing energy usage in residential areas with intermittent vehicle traffic.

8.5 Educational Projects

Serves as a valuable prototype for students learning embedded systems, sensor interfacing, and automation.

9. Advantages and Limitations

9.1 Advantages

- Energy Efficient — lights activate only when required
- Automatic Operation — no manual intervention needed
- Low Cost — uses widely available Arduino components
- Easy to Implement — straightforward circuit and code
- Real-Time Monitoring — LCD provides instant system status
- Expandable Design — easily upgradeable with more sensors
- Improved Road Safety — instant illumination on vehicle detection

9.2 Limitations

- Limited detection range of the HC-SR04 sensor (~4 meters max).
- Requires continuous power supply — no battery backup built in.
- Performance may be affected by adverse environmental conditions.
- Prototype uses LEDs instead of actual high-power street lights.

10.Future scope

10.1 Solar Integration

Use solar panels and rechargeable batteries to enable completely off-grid operation, making the system sustainable and cost-effective for remote areas.

10.2 IoT Monitoring

Monitor street lights remotely using ESP32 and Wi-Fi, enabling real-time dashboards and fault detection via mobile applications.

10.3 Motion Sensors

Integrate PIR sensors for improved detection of pedestrians and cyclists, not just vehicles.

10.4 Energy Monitoring

Add current and power measurement modules to track energy savings and generate reports.

10.5 Smart City Integration

Connect to centralized smart city infrastructure for coordinated traffic management and predictive lighting.

11.Conclusion

The Smart Street Light System with Vehicle Detection and LCD Monitoring successfully demonstrates the practical implementation of intelligent lighting automation using Arduino UNO. The project combines LDR-based day/night detection, ultrasonic vehicle detection, multiple street light control, delay-based operation, and LCD monitoring into a single embedded system.

The system effectively reduces unnecessary power consumption by operating street lights only when required while maintaining road safety through automatic vehicle-based illumination. The project provides valuable experience in embedded programming, sensor interfacing, automation logic, and real-time monitoring.

The developed prototype demonstrates the potential of intelligent street lighting systems and can be further expanded into a professional smart city solution through IoT connectivity, solar power integration, and advanced monitoring capabilities.

12.Reference

1. Arduino Official Documentation – <https://www.arduino.cc/reference/en/>
2. Aduino Uno datasheet - <https://docs.arduino.cc/resources/datasheets/A000066-datasheet.pdf>
3. 16×2 LCD Display Datasheet